



The humble PC

of operations. For a long while, these processors were enough to handle the workloads thrown at them. As time passed, these workloads became diverse and the processors started to evolve. Additional instruction sets were introduced, some of which were incrementally better than their predecessors while others opened up new avenues. The end result was that these processors started to become more complex.

Along the way, certain applications became more demanding. As a result, a separate co-processor had to be created to handle these workloads. The graphics card in consumer markets and AI accelerators in the enterprise market were born out of such needs. Over time, the design of these co-processors became efficient enough that they could be incorporated within the CPU itself. Which is why we have iGPUs (Integrated Graphics Processing Units) and elements for AI acceleration within processors. Intel's Iris Xe-LP is a good example of that. It's decently powerful to run popular video games while also being useful for AI features in applications such as Adobe Photoshop.

Does this mean that convergence would lead to the death of specialised processors? Of course not. Accelerators exist for a very good reason, they're great for han-

dling large volumes of operations that they were designed for. And with more applications and services heading into the cloud, the need for specialised accelerators will always remain. Aside from that, there are new features introduced into existing accelerators that ensure that these components remain discrete. Hardware-accelerated ray-tracing is one such feature. Video games were capable of producing high-fidelity global illumination before hardware-accelerated ray-tracing was introduced. And the GPUs were also evolving to become energy efficient. The moment hardware-accelerated ray-tracing was introduced, GPUs suddenly grew larger in size, started consuming oodles of power and gave off a lot of heat. These three factors were more than enough to stave off the eventual convergence that's expected for GPUs and the same applies to every piece of technology.

So while cut-down versions of powerful GPUs are integrated into CPUs, the GPUs themselves will always have a place. At least until we get to a level which allows video games with hyperrealism to be rendered on integrated graphics. That's not happening anytime before the next 10 years. And in that time, we might well see another compute-intensive feature being introduced.

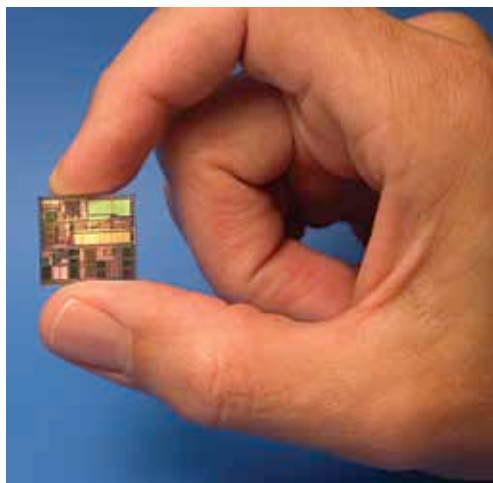
We will certainly see the desktop PC market diminish to less than half of what it is now in the next 15 years. That's how long it took for the desktop PC market to halve down

to its current level. Portable mobile form factors such as notebooks have already become quite powerful and in the coming 10 years, they will be immensely more capable and powerful. Current content creation machines will continue to retain their utility. We're currently using those to render 1080p and 4K video workloads and we'll just be upgrading to rendering 8K and some form of augmented reality content.

MOORE'S LAW

Not a law but more of an observation, Moore's Law states that the number of transistors within an integrated circuit will double every two years. Moore's Law has been plateauing for a while now. Now before you jump up and down and throw process node nomenclatures such as 14nm, 12nm, 7nm and everything that comes after it in our faces, let us repeat what we've said before in many articles. These terms are simply marketing monikers and the actual transistors aren't quite that small. Take TSMC for example, their 10nm process node had a density of 52.5 million transistors per square millimeters, followed by their 7nm which has a density of 96.5 million per mm² and 5nm has 173 million per mm². These nodes were launched at roughly two year intervals and we can see that the transistor density is growing by a factor of 1.7-to-1.8x. And this will continue into 3nm, 2nm and a little beyond that. IBM has already announced that it has successfully made a prototype of a chip based on 2nm which hosts 333 million transistors per mm².

After 2nm, it becomes immensely difficult to shrink transistors. At that level, these transistors are merely a few atoms thick. A 3nm film of silicon is just 15 atoms thick so you can imagine what we're dealing with. Again, just to clarify, this example was about an actual 3nm thick transistor film and not the kind of transistors that are built on a 3nm process. The latter is way thicker. So you're fighting a losing battle against physics. Of course, we're talking about silicon-based transistors. Switching to a different material



Convergence isn't so easy